Guest Editorial Enhancing Edge-Based Consumer Electronics Devices and IoT Ecosystem With Post-Quantum Cryptosystem: Security Challenges and Solutions

I. Introduction

THE conventional Internet of Things (IoT) architectures typically transmit a substantial amount of data to centralized cloud systems for processing that could strain bandwidth and introduce latency problems. Edge/fog computing model may shift an extensive portion of data processing to edge nodes at the network periphery. This local processing approach mitigates reliance on central cloud datacenters by forwarding only essential data from consumer IoT or edge computing (EC) devices to these datacenters. EC devices store and process sensitive data in a distributed manner that could expose it to a range of various threats. We can witness numerous edge-based intelligent security solutions designed for classical comparchitectures. Recent progress in the quantum computing paradigm severely imperils the widely adopted classic public-key cryptosystems as well as the enciphered data employing those algorithms.

Quantum computers can be used to efficiently solve mathematical problems that are considered computationally intractable for classical systems, intractable on the classical computing paradigm—thereby compromising widely adopted public-key cryptographic systems. The integration of post-quantum cryptography (PQC) into EC-assisted consumer IoT (EACI) environments presents a promising path for enhancing the security of consumer data by ensuring resilience against the quantum-aided adversaries. Most of the consumer IoT devices operate over extended lifespans that could last for decades—the need for long-term data safety becomes more challenging in the face of future quantum threats due to the quantum revolution.

To address these emerging challenges, this special issue explores quality research papers on PQC algorithms, protocols, theories, and hardware to curb emerging security threats for edge/fog-enabled EACI frameworks following the forthcoming quantum computing developments.

II. Articles Included in this Special Section

This Special Section of IEEE Transaction on Consumer Electronics comprises various aspects of enhancing the security of edge-based consumer electronics devices and the IoT ecosystem engaging post-quantum cryptography algorithms and protocols. With the tremendous advancements in quantum computing, this Special Section is timely, as it has drawn a large number of submissions. Out of these submissions, 12 high-quality research papers have been accepted, which explore the security aspects of the integration of PQC solutions with existing cloud, edge, and fog computing hardware technologies.

In [A1], Sharma and Rani strengthen the resistance of IoT to classical and quantum-based cyber attacks by integrating deep learning (DL) with PQC algorithms. For the provision of real-time anomaly detection and response expediency, it suggests a hybrid lightweight security framework with the combination of optimized PQC algorithms and a DL-assisted intrusion detection system (IDS) for deployment in a resource-constrained edge-based IoT ecosystem. The proposed methodology ensures secure data exchanges linked to quantum threats against IoT systems with the help of continuous learning through IDS models for adapting to novel attack vectors.

In [A2], Badar et al. suggested a novel secure communication paradigm for EC-enabled consumer devices and IoT-based smart grid ecosystem by integrating PQC for ensuring resilient data integrity and access control against the classical as well as quantum-assisted threats. The suggested framework uses a lightweight consortium blockchain with the use of the hyperledger fabric framework for maintaining tamper-proof records of authentication logs and energy-based transactions. For this purpose, regional edge gateways (REGs) are introduced to execute local crypto-operations and minimize computational requirements on resource-deficient IoT nodes. In addition, an AI-driven real-time IDS promotes the resilience of the suggested framework.

In [A3], Huang and Sharma suggest a lightweight group signcryption authentication scheme by using lattice cryptography to meet the increasing security requirements of resource resource-constrained edge-based IoT ecosystems. The scheme improves the key generation efficiency with the use of an optimized trapdoor diagonal matrix, supporting simultaneous signing as well as encryption through signcryption in a single logical step, and the leftover hash-based lemma. The proposed scheme introduces a low-interaction access control technique for group-based access authentication.

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In [A4], Ahmed and Anisi suggested a post-quantum secure federated learning framework for cross-domain vehicle-to-grid (V2G) authentication, since the reliance of existing cross-domain V2G access control mechanisms on centralized authorities may lead to a single point of failure and computational bottlenecks. This paper presents a lightweight quantum-resistant authentication protocol (LQAP) with the use of a hybrid extended Merkle signature scheme (XMSS) and a lightweight Merkle signature (LMS) system. A decentralized hierarchical federated learning (HFL) architecture at the edge layer supports distributed anomaly detection and secure hardware-assisted entity identification using physically unclonable functions (PUFs), without reliance on centralized data aggregation.

In [A5], Bagchi et al. suggested a ring-LWE-based multi-authority CP-ABE (MA-CP-ABE) protocol for addressing privacy and single-point failure problems in blockchain-based edge-IoT aided healthcare systems. The security of the protocol is enhanced against quantum threats with threshold secret-sharing by Shamir and the Lagrange interpolation formula. The protocol was implemented for CE-based IoT-enabled smart healthcare applications employing blockchain technology as a secure storage.

In [A6], Aneesh Kumar et al. contribute a hybrid architecture with the combination of quantum key distribution (QKD), chaos-based encryption, and PQC for securing multimedia data in IoT-oriented consumer electronic networks. The intent was to address the challenges posed by quantum technology to classical image encryption algorithms. This integrates a 2-D-logistic sine coupling map (2D-LSCM) with the keys generated from QKD and the NTRU algorithm. In this manner, the scheme largely enhances image processing speeds, the performance of the chaotic map, confidentiality, and sensitivity values.

In [A7], Farouk et al. suggest a quantum blockchain framework for scalable and secure integration with 6G networks. The scheme employs temporally entangled GHZ states, promoting inherent quantum encoding of block linkage and integrity, and a four-qubit blockchain data structure implemented with the use of IBM Q quantum processors. The system largely ensures quantum-resilient block integrity and tamper resistance against quantum threats, supporting useful 6G consumer applications like telemedicine, IoT authentication, and decentralized finance.

In [A8], Seno et al. suggest a novel blind signature scheme (CDBS) for securing consumer electronic IoT (CIoT) applications against quantum attacks. The motivation was to overcome the challenges of traditional blink signature schemes. The proposed CDBS is based on NIST-standardized CRYSTALS-Dilithium and incorporates blinding as well as unblinding steps—supporting enhanced privacy for applications related to anonymous payments and privacy-focused e-ticketing.

In [A9], Mahmood et al. proposed APPLETA, an efficient trust management framework for resource-constrained IoT devices. It calculates trust values based on application transactions and authentication behaviors without additional overhead. Moreover, APPLETA effectively detects and isolates malicious nodes, exhibiting strong resistance to good and bad-mouthing threats in simulations.

In [A10], Ullah et al. introduced Q-P2FL, a quantum-based privacy-preserving federated learning technique for the detection of adversarial attacks on consumer edge devices. The scheme combines quantum intelligence, additive homomorphic encryption, and vision transformer (ViT)-based feature extraction to ensure data privacy and model security. This approach largely suggests a convincing solution for safeguarding consumer edge devices from adversarial threats and improving their privacy-preserving capabilities.

In [A11], Cao et al. design a physical layer security fog computing IoT (PSFC-IoT) system, which ensures physical layer security (PLS) against multiple eavesdroppers in a fog computing IoT system. A joint resource allocation method based on the post-quantum beetle optimization algorithm (QBOA) is employed for optimal resource allocation. Simulations confirm the enhanced secrecy rates under varying conditions while meeting optimal latency requirements for consumer electronics such as wearables, linked devices, and smart home devices.

To overcome the Kyber protocol's requirement of prior authentication, in [A12], Shahidinejad et al. present a NIST-compliant quantum-resistant password-authenticated key exchange (PAKE) for mobile devices. In this way, the scheme manages to counter quantum threats in the conventional key exchange framework. Leveraging CRYSTALS-Kyber encryption, it supports anonymity, mutual authentication, and perfect forward secrecy. The hardware-based implementation and formal analysis validate the scheme's security and efficiency. In comparison with Kyber's protocol, it manages to cut communication and computation overheads.

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APPENDIX: RELATED ARTICLES

[A1] A. Sharma and S. Rani, "Post-quantum cryptography (PQC) for IoT-consumer electronics devices integrated with deep learning," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 4925–4933, May 2025.

[A2] H. M. S. Badar, S. Ahmed, N. I. Kajla, G. Fan, and C. Zhang, "Q-BLAISE: Quantum-resilient blockchain and AI-enhanced security protocol for smart grid IoT," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 4959–4971, May 2025.

[A3] R. Huang and A. Sharma, "Post-quantum verifiable decryption scheme for Internet of Thing-based consumer electronic edge device," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 4903–4913, May 2025.

[A4] S. Ahmed and M. H. Anisi, "A post-quantum secure federated learning framework for cross-domain V2G authentication," *IEEE Trans. Consum. Electron.*, early access, Jun. 16, 2025, doi: 10.1109/TCE.2025.3580338.

[A5] P. Bagchi, A. Bisht, A. K. Das, N. Saxena, and M. S. Hossain, "Designing quantum-safe lattice-based multi-authority CP-ABE scheme for blockchain-enabled IoT-based consumer healthcare electronics," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 4983–4994, May 2025.

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[A7] A. Farouk, B. K. Behera, and E. A. Ahmed, "Design and implement a quantum blockchain framework to secure 6G communication for consumer applications," *IEEE Trans. Consum. Electron.*, early access, Jun. 17, 2025, doi: 10.1109/TCE.2025.3580583.

[A8] M. E. Seno et al., "Post quantum-resistant blind signature scheme for consumer Internet of Things security," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 4949–4958, May 2025.

[A9] S. Mahmood, M. Gohar, S.-J. Koh, M. U. Tariq, and A. Ghani, "Application level trust authority (APPLETA) for resource-constrained edge devices in IoT and 6G," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 4934–4948, May 2025.

[A10] F. Ullah, N. Mohammad, L. Mostarda, D. Cacciagrano, and Y. Zhao, "Q-P2FL: Quantum-enhanced federated edge intelligence for privacy-preserving adversarial attack detection on consumer edge devices," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 4914–4924, May 2025.

[A11] Z. Cao, B. Geng, and D. Yu, "Consumer technology in focus: Post-quantum security for fog computing IoT systems against eavesdropping," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 4972–4982, May 2025.

[A12] A. Shahidinejad, J. Abawajy, and S. Huda, "Quantum-proof anonymous key exchange with perfect forward secrecy for mobile devices," *IEEE Trans. Consum. Electron.*, early access, Jun. 24, 2025, doi: 10.1109/TCE.2025.3582742.

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